AUSELIA

INTERNAL TECHNICAL DOCUMENT

ADS7057 AFE test board: bring-up guide

Seven steps from the first power-up to a head on the plant

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VERSION 0.1

STATUS Draft for the first boards

1 About this guide

This guide takes the first assembled boards from the box to a measured result. It follows the order of the signal: power first, then the chain, then the listener, then the SPI link, then the sensor head, and last the head on the plant.

Each step has the same parts: a goal, what to do, a table of what you should see, a green *Pass* box and an amber *If not* box. Every "expected" number comes from the ngspice simulation of the *as-built* netlist (`sim/netlist_verify.py`) or from the datasheets. No number here has been measured on a real board yet: your readings are the first.

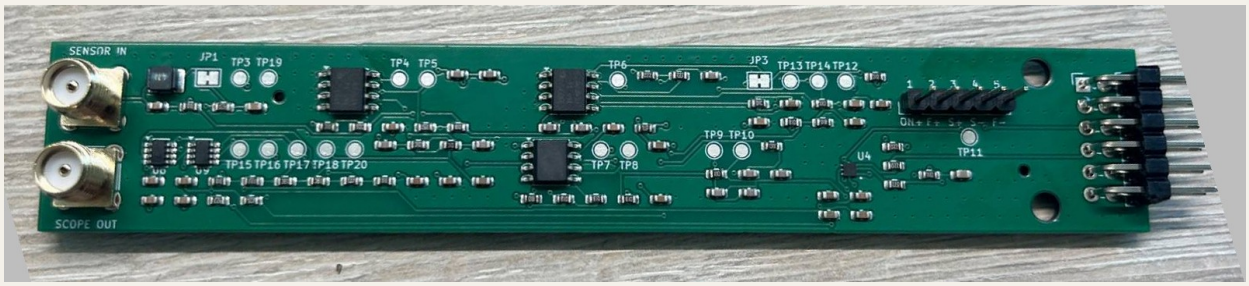


Figure 1. The AFE Test Board v1, top side, as delivered. SENSOR IN (J1) and SCOPE OUT (J4) are on the left, the power header J7 and the PMOD J3 on the right.

1.1 What you need

Item	Use
Keysight B2901BL (SMU)	Powers the board through J7 and reads the supply current, four-wire
Function generator or Analog Discovery 2	Bursts of 250 kHz, 50 Ω output, 10 cycles
Oscilloscope, ≥ 250 MS/s	Probes on the test points; 1 MΩ input
Multimeter	Resistances and DC voltages
ZedBoard with a program that reads the ADS7057	SPI link and listener lines (Step 4)
10 pF C0G capacitor (±1 %) and 1600 pF C0G capacitor	Charge injection and dummy sensor for the head
SMA cables, one jumper for J7, a knife, a soldering iron	Cut and close JP1 and JP3
Magnifier or microscope	Inspection of U4, U6, U9

Before you connect a generator to J1, cut JP1. With JP1 closed the centre pin of J1 carries about 3.3 V DC for the sensor head, and the generator would feed it. Cut JP1 with a knife for the generator tests (Steps 2 and 3). Close it again with a drop of solder for the head (Steps 5 to 7).
Work on an ESD mat or touch a grounded surface first. The ADS7057 and the two TLV7022 are small parts with fine pitch.

2 Know the board

Figure 2 shows where each part and test point sits. The red dot on an IC is its pin 1. Table 1 lists every test point with the stage it measures.

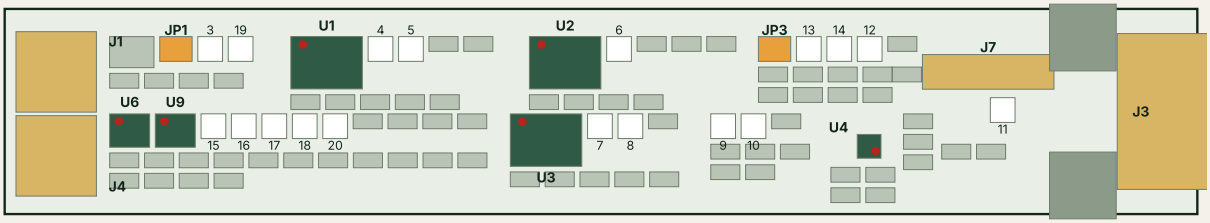


Figure 2. Top view with every part. Green: ICs with the red pin-1 dot. Gold: connectors. Orange: solder jumpers JP1 and JP3. White: test points (the number after TP).

Table 1. Test points of the main board. “SIG_IN” is the input after the bias-T.

TP	Net	Shows	TP	Net	Shows
TP3	SIG_IN	input	TP12	AVDD	ADC reference
TP4	HPF_OUT	high-pass	TP13	VMID5	1.65 V mid-rail
TP5	GAIN_OUT	gain stage	TP14	VOCM	driver centre
TP6	LPF_OUT	low-pass	TP15	LST	listener signal
TP7	DRV_OUTP	driver +	TP16	WAKE_N	small window
TP8	DRV_OUTN	driver −	TP17	BIG_N	large window
TP9	AINP	ADC input +	TP18	REFL	listener centre
TP10	AINM	ADC input −	TP19, TP20	GND	scope ground
TP11	+3V3	board rail			

2.1 The signal path in one picture

Figure 3 shows how big the signal is at each test point for a sine at 250 kHz. The chain gain is about 4.1 to the ADC, and the driver splits the signal into two halves of opposite sign around 1.65 V.

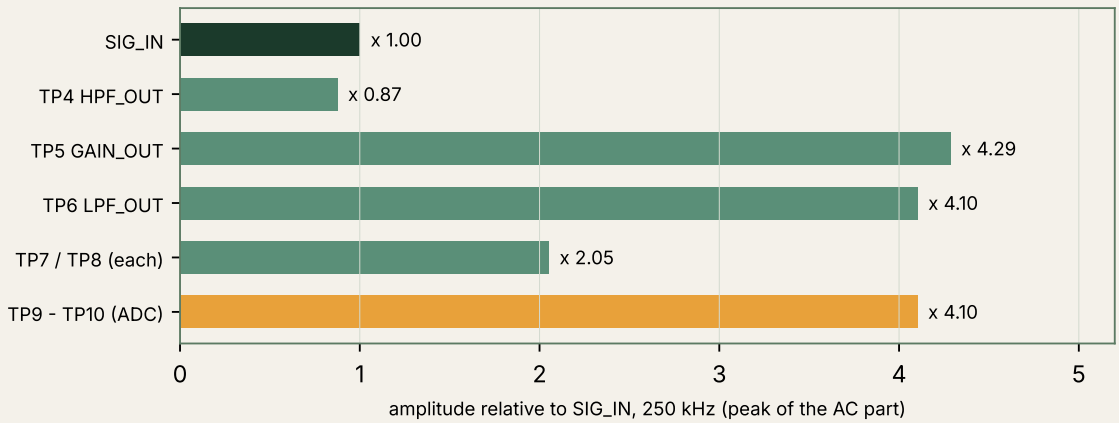


Figure 3. Expected amplitude at each test point at 250 kHz, relative to the amplitude at SIG_IN (ngspice, as-built netlist).

STEP 0 Look first, power off

Goal: find a wrong part, a short or a solder bridge before any current flows.

1. Look at the top side under the magnifier. Compare with Figure 2 and the photo on page 2.
2. Check the pin-1 end of every IC. The part has a dot or a bevelled edge, and the board has a triangle on the silkscreen.
3. Look at U4 (ADS7057, 1.5 mm, eight pads), U6 and U9 (TLV7022, SOT-23-8). Look for solder bridges between pads.
4. Check that the pins of J3 and J7 are straight and that J3 is flush with the board edge.
5. With the multimeter, measure the resistance between TP11 (+3V3) and TP19 (GND). Wait until the reading stops changing.

Table 2. Pin-1 positions, seen from the top with SENSOR IN on the left.

Ref	Part	Package	Pin 1
U1, U2, U3	OPA2365, OPA2365, THS4531	SOIC-8	top left; the silkscreen triangle is next to it
U4	ADS7057	X2QFN-8	lower right of the part
U6, U9	TLV7022	SOT-23-8	top left
U8 (head)	OPA365	SOT-23-5	right side, lower pin, triangle on the silkscreen

TP11 to TP19: about 2 kΩ (ESTIMATE: the 1 kΩ pair of the VOCM divider, the 20 kΩ of the VMID5 divider and the 2.1 MΩ ladder in parallel). Anything above 1 kΩ is acceptable. No bridges, all pin-1 ends as in the table.

Below 500 Ω is a short: do not power on. Look for a bridge at U4, U6, U9, a turned IC, or a bridged C0G capacitor. A reading that climbs slowly is only the capacitors charging.

STEP 1 Power up with the SMU

Goal: the board draws the right current and every rail is at its value.

The board takes 3.3 V on the power header J7. The ZedBoard stays unplugged for this step.

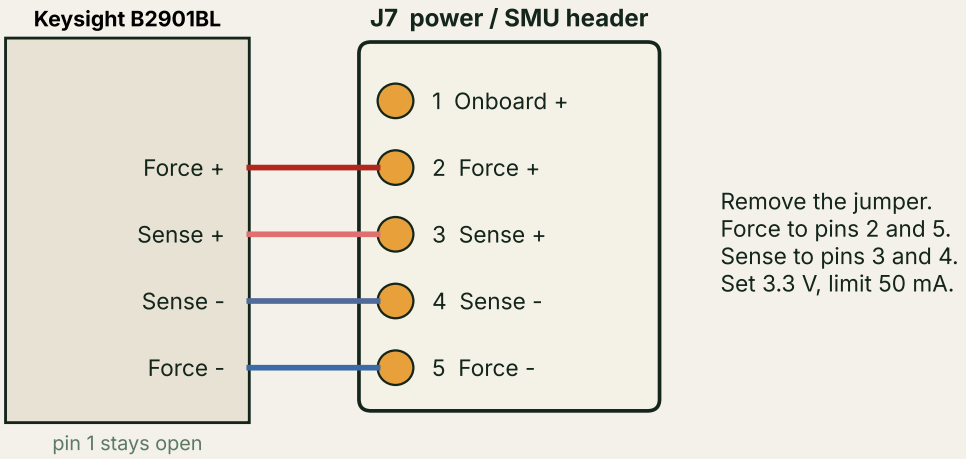


Figure 4. Four-wire connection of the B2901BL to J7. Pins 2 and 3 meet at the board rail, pins 4 and 5 are ground.

1. Remove the jumper from J7. Leave the PMOD unplugged and JP1 as it is (closed).
2. Set the SMU to voltage source, 3.3 V, current limit 50 mA, remote sense on. Output off.
3. Connect Force to pins 2 and 5 and Sense to pins 3 and 4 of J7.
4. Switch the output on. Read the current after one second.
5. Measure each node of Table 3 against GND (TP19).

Table 3. Expected DC values (ngspice, all links closed, head not connected unless said). Tolerance: ±30 mV for the 3.3 V nodes, ±50 mV for the others.

Node	Test point	Expected	Note
+3V3	TP11	3.30 V	the SMU value
AVDD	TP12	3.27 V	after the 10 Ω and 10 µF filter
VMID5	TP13	1.65 V	needs 25 ms after power-up
VOCM	TP14	1.64 V	half of AVDD
LST	TP15	1.64 V	rests at REFL
WAKE_N, BIG_N	TP16, TP17	3.30 V	open drain, pulled up
REFL	TP18	1.64 V	middle of the ladder
Supply current	SMU	20 to 22 mA	without the head; about 25 mA with it

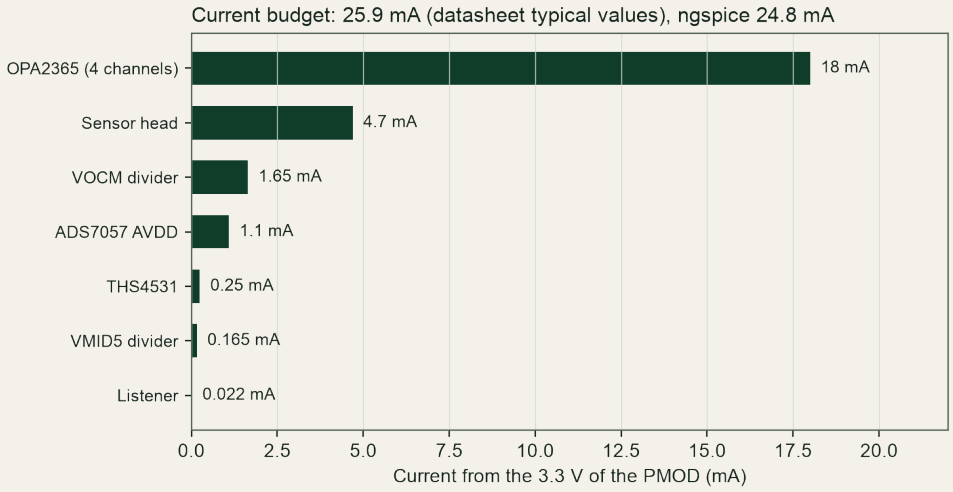


Figure 5. Where the current goes (datasheet typical values, estimate). The four op-amp channels are the largest part.

Split the current with JP3

JP3 joins the capture rail (U1, U2, U3 and AVDD) to the 3.3 V rail. Power off, cut JP3 with the knife and connect the SMU to the capture side instead, or measure with a second ammeter across the gap. Expected: about 20 to 22 mA on the capture rail and about 0.3 mA on the rest (listener and DVDD). Close JP3 with a drop of solder when you are done.

Current between 18 and 30 mA, all nodes in Table 3. Stop at once if the current passes 60 mA or if TP11 is off by more than 0.1 V.

Current too high: power off, open JP3 and measure each side. A high capture side points to U1, U2, U3 or a bridged capacitor. A high other side points to U6, U9 or U4. Voltage low at TP11: check the Sense wires. A wrong mid-rail: wait 25 ms, and then check R7, R8, R14 and R15.

Then with the ZedBoard. Power off. Put the jumper back on J7 pins 1 and 2 (this joins the PMOD 3.3 V to the board rail), and plug the board into a PMOD port. Never use the jumper and the SMU together.

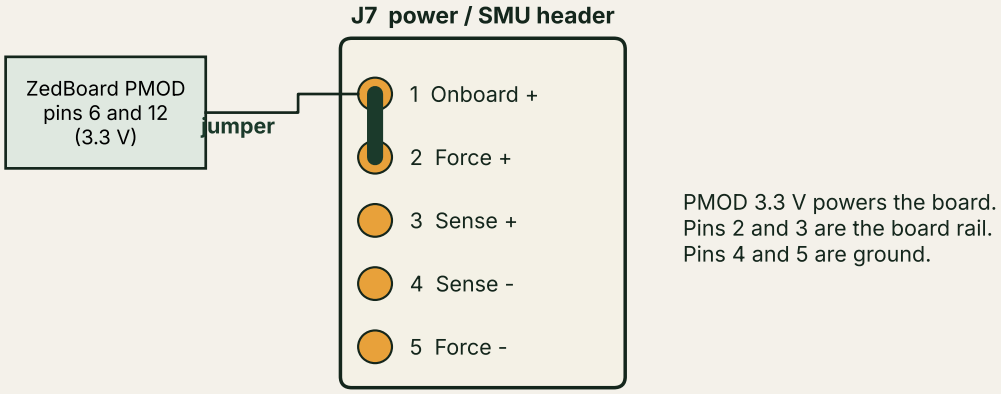


Figure 6. Powering the board from the ZedBoard PMOD: a jumper on J7 pins 1 and 2.

STEP 2 The signal chain

Goal: each stage has the gain and the band that the simulation gives.

Cut JP1 first (see page 2). Use a 50 Ω generator and keep the amplitude at SIG_IN under 300 mV peak, or the gain stage clips.

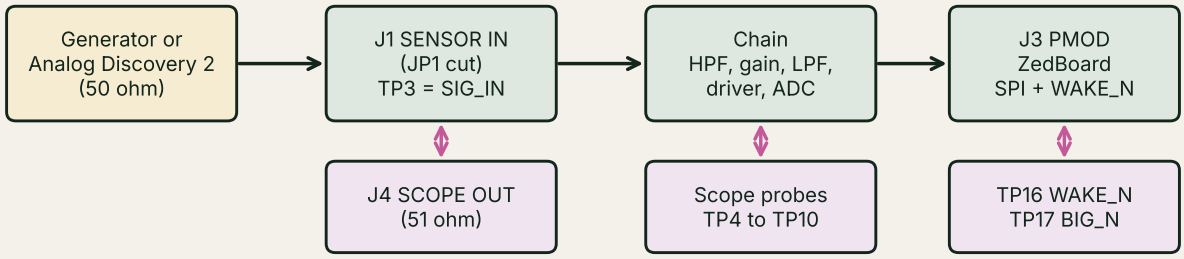


Figure 7. Test setup for Steps 2 to 4.

1. Power the board as in Step 1 (SMU or jumper).
2. Set the generator to 250 kHz, a burst of 10 cycles, repeated every 1 ms, 100 mV peak (200 mV peak to peak). Connect it to J1.
3. Trigger the scope on TP3 (SIG_IN). Measure the peak of the AC part at each test point of Table 4.
4. Look at the waveform of TP7 and TP8: they are opposite in phase around 1.65 V (Figure 8).
5. Replace the burst with a continuous sine of 100 mV peak and set it to 10, 30, 100, 250, 500 kHz, 1 and 2 MHz. Read TP6. Compare with Table 5.
6. Probe SCOPE OUT (J4) with a 1 MΩ input. It must match TP6 (51 Ω in series, the scope takes half or less if the input is 50 Ω).

Table 4. Expected peak at each test point for 100 mV peak at SIG_IN, 250 kHz. Limit: ±10 %.

Test point	Net	Peak (mV)	Gain
TP3	SIG_IN	100	1.00
TP4	HPF_OUT	87	0.87
TP5	GAIN_OUT	429	4.29
TP6	LPF_OUT	410	4.10
TP7, TP8	DRV_OUTP, DRV_OUTN	±205 each	opposite signs around 1.65 V
TP9, TP10	AINP, AINM	±205 each	same as the driver (10 Ω only)
TP15	LST	68	0.68 (listener filter)

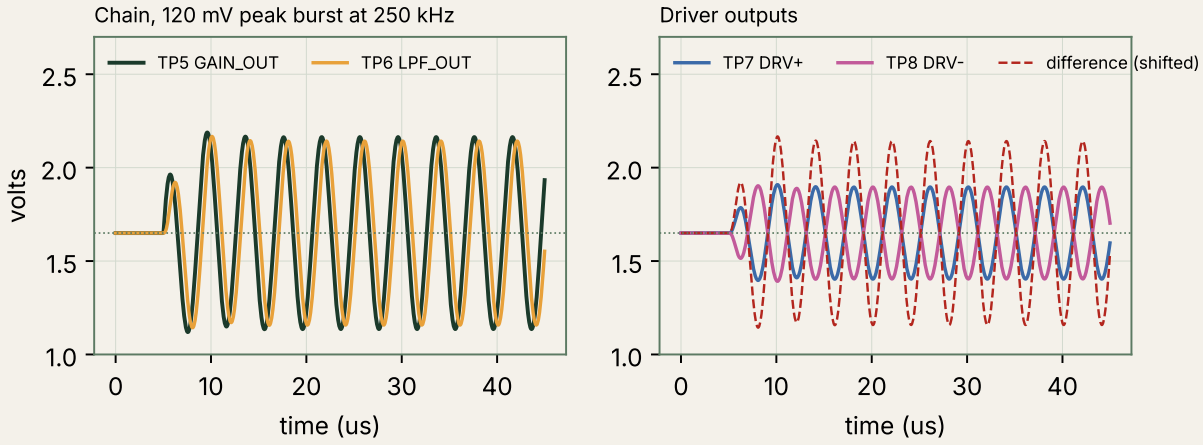


Figure 8. Expected waveforms for a burst of 120 mV peak (ngspice). The ADC sees the difference of TP7 and TP8.

Table 5. Expected response, relative to 250 kHz (ngspice, whole chain to the ADC). Limit: ±1.5 dB from 100 kHz to 500 kHz, ±3 dB outside.

f	10 kHz	30 kHz	100 kHz	250 kHz	500 kHz	1 MHz	2 MHz
dB	−40.5	−21.5	−3.1	0	−2.8	−12.1	−24.0

Peaks within 10 % of Table 4 and the sweep within 1.5 dB of Table 5 from 100 to 500 kHz.

Find the first stage that is wrong. If TP4 is wrong, check C1, C2, R1, R2 and U1 pin 1. If TP5 is wrong, check R3 (3.9 kΩ) and R4 (1 kΩ). If TP6 is wrong, check R5, R6, C4 to C6. If TP7 and TP8 are not opposite, check U3, R10 to R13 and VOCM at TP14. Large peaks at one end only point to a wrong capacitor value.

STEP 3 The listener: WAKE_N and BIG_N

Goal: the two window comparators fire near 35 mV and 111 mV at SIG_IN.

Keep the setup of Step 2 (JP1 cut). Probe TP15 (LST), TP16 (WAKE_N), TP17 (BIG_N) and TP18 (REFL).

1. Check that TP18 reads 1.65 V and that TP16 and TP17 read 3.3 V with no signal.
2. Send the 250 kHz burst of 10 cycles at 20, 60 and 150 mV peak at SIG_IN.

3. Note which line falls. A line stays low only for the length of the burst.
4. To find the exact thresholds, raise the amplitude in steps of 5 mV until WAKE_N falls, then until BIG_N falls.

Table 6. Expected lines (250 kHz, 10 cycles, JP1 cut, peak at SIG_IN). Each threshold is good to about ±12 mV.

Amplitude	WAKE_N (TP16)	BIG_N (TP17)
20 mV	stays high	stays high
60 mV	falls	stays high
150 mV	falls	falls
threshold	about 35 mV	about 111 mV

The three rows of the table, and thresholds within ±12 mV. The response time is about 260 ns.

WAKE_N always low: check TP18 (must be 1.65 V), then R39 to R41 and the 1 MΩ resistors R21 and R23. A threshold far off: check R22 and R41 (33 kΩ) and R39 and R40 (15 kΩ). No fall at all: check U6, U9 and the pull-ups R24, R42.

STEP 4 The SPI link and the ADC

Goal: the ZedBoard reads codes that follow the input.

You need a program in the ZedBoard fabric that drives CS, SCLK and reads SDO. The pins are in Table 7. There is no MOSI because the ADC only sends data.

Table 7. PMOD J3 (the board uses nine of the twelve pins).

Pin	Signal	Pin	Signal
1	CS	7	BIG_N
2	not used	8	not used
3	SDO	9	not used
4	SCLK	10	WAKE_N
5	GND	11	GND
6	3V3	12	3V3

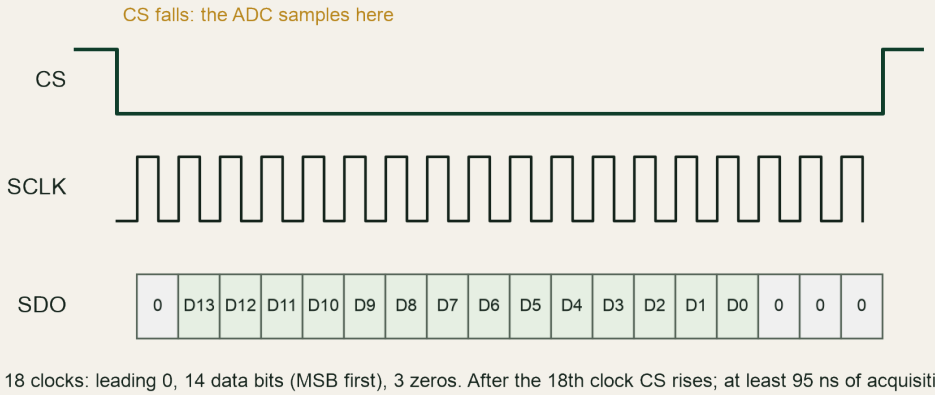


Figure 9. One frame: CS falls (the ADC samples), 18 clocks, then CS rises and the ADC acquires for at least 95 ns.

1. With no signal on J1, run SCLK at 10 MHz. Send one frame of 24 clocks first (offset calibration), and then frames of 18 clocks.
2. Check on the scope at the J3 pins: CS, SCLK and SDO as in the figure. The first frame returns zeros. Every later frame returns the sample of the frame before.
3. Read 4096 samples with no signal. Calculate the mean (offset) and the rms (noise) in codes. One code is 0.403 mV.
4. Send the 250 kHz sine of 100 mV peak at SIG_IN. Read 4096 samples and find the peak code.
5. Raise SCLK to 20 MHz and then to 25 MHz, and check for errors. If there are errors, move the sample edge of the host later.

Table 8. Expected codes.

Test	Expected	Limit
First frame (24 clocks)	all zeros	-
Offset with the input idle	about 0 codes	±10 codes
Noise with the input idle (no sensor)	about 1 code rms (ESTIMATE)	below 3 codes
Peak for 100 mV at SIG_IN, 250 kHz	±1018 codes	±10 %
Sample rate: 10 / 20 / 25 / 60 MHz	0.53 / 1.01 / 1.23 / 2.50 MSPS	-

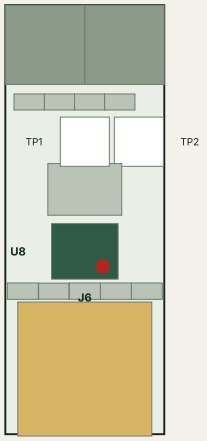
Codes follow the input in size and sign, SCLK at 25 MHz gives no errors, and the idle noise is below 3 codes.

All zeros always: read twice, since the data comes one frame late. SDO stuck: check +3V3 at U4 pin 4 and R18 to R20 (22 Ω). Wrong sign or a shifted word: the data has a leading zero, then 14 bits (D13 first, two's complement), then three zeros. A large offset: check VOCM at TP14 and VMID5 at TP13.

STEP 5 The sensor head alone

Goal: the charge amplifier turns charge into voltage with the gain Q / Cf.

The head (10 × 27 mm) is powered through the coax from SENSOR IN. Close JP1 on the main board for this step, and put the SMA cable between J6 of the head and J1.



Head connections

TP1 (+)	piezo + pad, the summing node
TP2 (−)	piezo − pad, head ground
J6	SMA to J1 of the main board
U8	OPA365, pin 1 as in Step 0

Figure 10. The sensor head. The two large holes at the top are the M2 mounting holes.

1. Wait for the head to settle. The reference divider (two 1 MΩ resistors and 10 µF) takes about 25 s.
2. Measure the DC at the head: the supply pad of C37 about 2.9 V, the reference pad of C35 about 1.45 V. Compare the supply current: about 4.3 mA more than in Step 1.
3. Inject charge: connect the generator through a 10 pF C0G capacitor to TP1, with the generator ground on TP2. Use a burst of 250 kHz and 10 cycles.
4. Read the peak at TP3 on the main board for each row of Table 9.
5. Put the 1600 pF C0G capacitor between TP1 and TP2 as a dummy sensor, and record TP6 with the input quiet. The noise gain rises to 35, as with the real sensor.

Table 9. Expected head output (Cf = 47 pF, Q = 10 pF × V, tolerance ±10 %).

Generator (peak)	Charge	Head output at TP3	TP6 (chain)	Listener
100 mV	1 pC	21 mV	87 mV	none
300 mV	3 pC	64 mV	262 mV	WAKE_N falls
1000 mV	10 pC	213 mV	874 mV	both fall

Outputs within 10 % of the table. A real event of 4 to 7 pC gives 85 to 149 mV: WAKE_N falls for all, and BIG_N falls above 5.2 pC. The noise at the ADC is below 2 codes with the dummy sensor (simulation: 0.2 codes).

No output: check that JP1 is closed and that the head DC is present. Wrong DC at the head: check U8 pin 1 and the 1 MΩ resistors R29 and R30. A gain error: check C34 (47 pF C0G) and R28 (1 MΩ). A very large noise with the dummy sensor: wait 25 s, then check the coax and ground.

STEP 6 Head and board together

Goal: one full chain from the head connector to the ZedBoard.

Connect the head to J1 with the coax (JP1 closed), power the board from the PMOD, and repeat the injection of Step 5 at 3 pC while the ZedBoard reads the ADC. The peak code must be about 262 mV × 1 / 0.403 mV ≈ 650 codes and WAKE_N must fall for the length of the burst.

STEP 7 The sensor on the plant

Goal: the first real recording.

Solder the piezo to the head, mount it on the plant, and record with a window long enough for several events. Compare the peak frequency (233 to 275 kHz in the old data) and the amplitude (4 to 7 pC) with the perfboard results. Note two numbers for the next design: the rms noise in codes, and the highest frequency present in an event.

3 Troubleshooting

Work from the top of Figure 11: find the first question that you answer “no”, and look only at that stage.

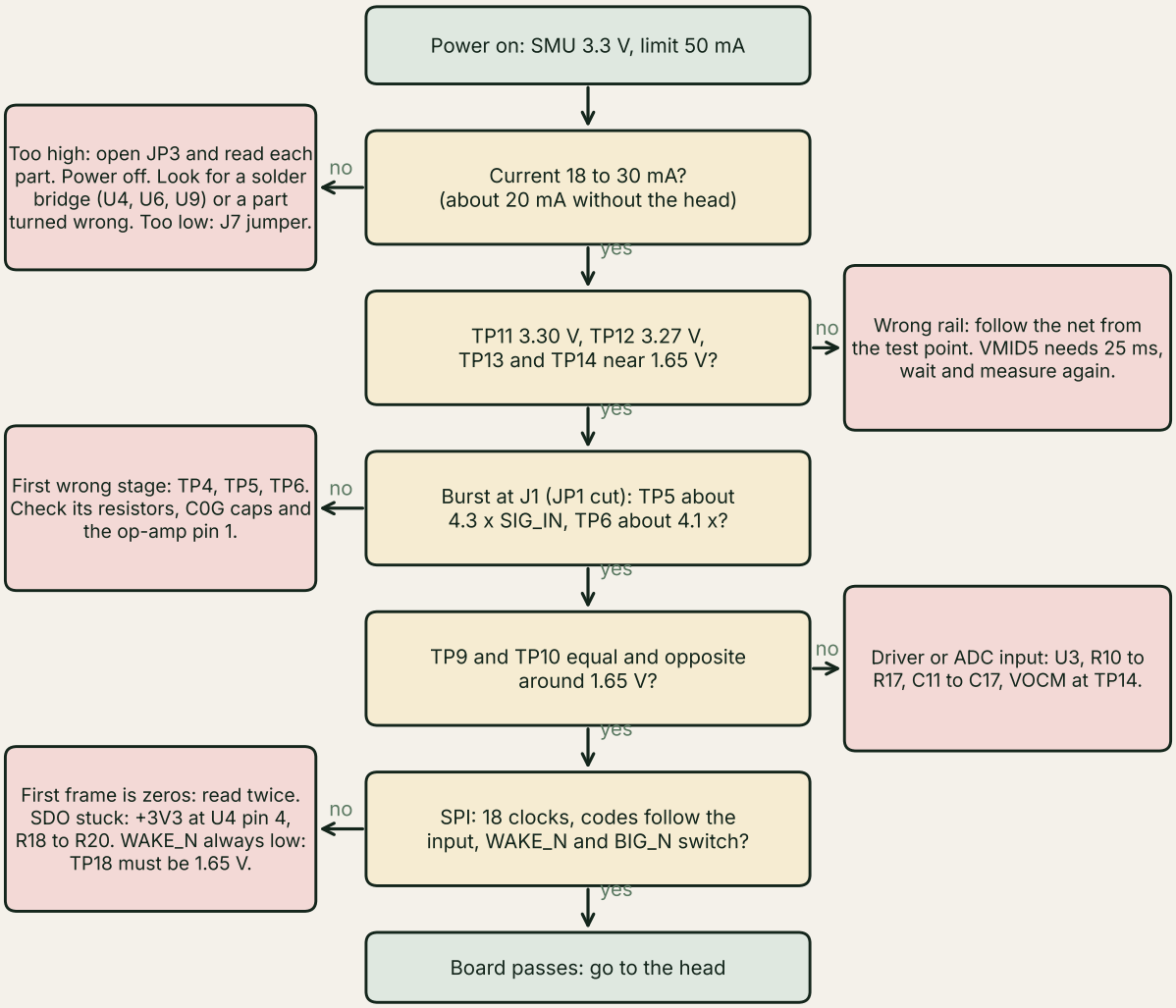


Figure 11. Decision chart for the main board.

4 Record sheet

Fill in the readings. Board serial: _____ Date: _____ Power: ☐ SMU ☐ PMOD

Check	Expected	Measured	Limit	OK
TP11 to TP19 resistance	about 2 kΩ		$>$ 1 kΩ	
Supply current (no head)	20 to 22 mA		18 to 30 mA	
+3V3 (TP11)	3.30 V		±30 mV	
AVDD (TP12)	3.27 V		±50 mV	
VMID5 (TP13)	1.65 V		±50 mV	
VOCM (TP14)	1.64 V		±50 mV	
REFL (TP18)	1.64 V		±50 mV	
TP5 / TP3 at 250 kHz	4.29		±10 %	
TP6 / TP3 at 250 kHz	4.10		±10 %	
TP7, TP8 (each, 100 mV in)	±205 mV		±10 %	
Sweep at 100 / 500 kHz (dB)	-3.1 / -2.8		±1.5 dB	
WAKE_N threshold	35 mV		±12 mV	
BIG_N threshold	111 mV		±12 mV	
ADC offset (codes)	0		±10	
ADC noise, input idle (codes rms)	about 1		$<$ 3	
Peak code for 100 mV at 250 kHz	1018		±10 %	
Head: DC at C37 / C35	2.9 V / 1.45 V		±0.1 V	
Head: 3 pC at TP3	64 mV		±10 %	

Notes